Week 2 Internship Report

Network Scanning & Enumeration Using Nmap

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Week: 2
Tool Used: Nmap (Network Mapper)
Environment: Kali Linux, Metasploitable2

1. Introduction

During the second week of my Red Teaming internship at Cyberster, I focused on **network scanning and enumeration techniques** using the tool **Nmap**.

Network scanning is an essential phase of penetration testing where security professionals identify:

- Active hosts
- Open ports
- Running services
- Service versions
- Potential vulnerabilities

The objective of this week was to gain hands-on experience with **Nmap scanning techniques, service detection, scripting engine usage, and stealth techniques**.

Two environments were used for testing:

1. **Resolved subdomains of hackedthissite.org**
2. **Metasploitable2 vulnerable machine**

2. Installing Nmap

The first step was installing Nmap in the Kali Linux environment.

Command used:

```
sudo apt install nmap
```

Nmap is a powerful open-source tool used for **network discovery and security auditing**.

3. Task 1 — TCP and UDP Port Scanning

The first task involved scanning **resolved subdomains** stored in a file named:

```
resolved.txt
```

This file contained **33 active subdomains** whose DNS records were successfully resolved.

```
4-v3stage-cdn.hackthissite.org
1-v3dev-cdn.hackthissite.org
irc-hub.hackthissite.org
4-v3dev-cdn.hackthissite.org
irc-lille.hackthissite.org
mirror.hackthissite.org
wolf.irc.hackthissite.org
v3stage.hackthissite.org
psrp.hackthissite.org
hub.irc.hackthissite.org
h5ai.hackthissite.org
v3dev-cdn.hackthissite.org
irc.hackthissite.org
0-v3dev-cdn.hackthissite.org
legal.hackthissite.org
v3dev.hackthissite.org
2-v3stage-cdn.hackthissite.org
3-v3dev-cdn.hackthissite.org
status.hackthissite.org
irc-wolf.hackthissite.org
1-v3stage-cdn.hackthissite.org
forums.hackthissite.org
v3stage-cdn.hackthissite.org
lille.irc.hackthissite.org
3-v3stage-cdn.hackthissite.org
2-v3dev-cdn.hackthissite.org
hp.hackthissite.org
stats.hackthissite.org
www.irc.hackthissite.org

┌──(kali㉿kali)-[~]
└─$ wc -l resolved.txt
33 resolved.txt
```

3.1 TCP SYN Scan

The TCP SYN scan is one of the most commonly used scanning techniques to identify open ports.

Command used:

```
nmap -sT -iL resolved.txt
```

```
┌──(kali㉿kali)-[~]
└─$ cat tcp_scan.txt
# Nmap 7.95 scan initiated Wed Mar 11 04:47:33 2026 as: /usr/lib/nmap/nmap --privil
Nmap scan report for www.hackthissite.org (137.74.187.101)
Host is up (0.14s latency).
Other addresses for www.hackthissite.org (not scanned): 137.74.187.102 137.74.187.1
rDNS record for 137.74.187.101: hackthissite.org
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE
80/tcp  open  http
443/tcp open  https

Nmap scan report for 0-v3stage-cdn.hackthissite.org (198.148.81.146)
Host is up (0.0019s latency).
rDNS record for 198.148.81.146: customer.sharktech.net
All 1000 scanned ports on 0-v3stage-cdn.hackthissite.org (198.148.81.146) are in ig
Not shown: 1000 filtered tcp ports (no-response)
```

Explanation

- **-sT** → Performs a TCP Connect scan
- **-iL** → Reads targets from a file

Result

The scan identified active hosts and detected open ports on several resolved subdomains.

3.2 TCP Scan with Output File

To save scan results for documentation and analysis, the scan output was stored in a file.

Command used:

```
nmap -sS -iL resolved.txt -oN scan.txt
```

Explanation

- **-sS** → TCP SYN stealth scan
- **-iL** → Input list of targets
- **-oN** → Saves output in normal format

Result

The scan results were successfully saved into:

`scan.txt`

3.3 UDP Port Scanning

UDP scans are useful for discovering services that run on **UDP ports**, such as DNS, SNMP, and DHCP.

Command used:

```
sudo nmap -sU --top-ports 100 -iL resolved.txt
```

Explanation

- **-sU** → UDP scan
- **--top-ports 100** → Scans the top 100 most common ports
- **-iL** → Reads target hosts from file

Result

The scan identified UDP services running on several hosts.

```
┌──(kali㉿kali)-[~]
└─$ cat udp.txt
# Nmap 7.95 scan initiated Wed Mar 11 13:29:15 2026 as: /usr/lib/nmap/nmap --pr
Nmap scan report for www.hackthissite.org (137.74.187.102)
Host is up (0.14s latency).
Other addresses for www.hackthissite.org (not scanned): 137.74.187.100 137.74.1
rDNS record for 137.74.187.102: hackthissite.org
All 100 scanned ports on www.hackthissite.org (137.74.187.102) are in ignored s
Not shown: 100 open|filtered udp ports (no-response)

Nmap scan report for 0-v3stage-cdn.hackthissite.org (198.148.81.146)
Host is up (0.0011s latency).
rDNS record for 198.148.81.146: customer.sharktech.net
All 100 scanned ports on 0-v3stage-cdn.hackthissite.org (198.148.81.146) are in
Not shown: 100 open|filtered udp ports (no-response)

Nmap scan report for qdb.hackthissite.org (137.74.187.153)
Host is up (0.00062s latency).
rDNS record for 137.74.187.153: www.irc.hackthissite.org
All 100 scanned ports on qdb.hackthissite.org (137.74.187.153) are in ignored s
Not shown: 100 open|filtered udp ports (no-response)

Nmap scan report for ctf.hackthissite.org (185.199.108.153)
Host is up (0.0040s latency).
Other addresses for ctf.hackthissite.org (not scanned): 185.199.109.153 185.199
rDNS record for 185.199.108.153: cdn-185-199-108-153.github.com
```

Aggressive scanning performs **advanced detection techniques** including:

- OS detection
- Service version detection
- Script scanning
- Traceroute

Due to restrictions on the external target website, the aggressive scan was performed on a **Metasploitable2 virtual machine**.

Command used:

```
sudo nmap -A -T4 -iL resolved.txt
```

Explanation

- **-A** → Aggressive scan (OS detection, scripts, traceroute)
- **-T4** → Faster scan timing
- **-iL** → Input file containing targets

Result

The scan detected multiple services and vulnerabilities running on the **Metasploitable2 server**, including various outdated services.

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -A 192.168.1.9 -oN aggressive.txt
sudo] password for kali:
tarting Nmap 7.95 ( https://nmap.org ) at 2026-03-14 09:35 EDT
map scan report for 192.168.1.9
ost is up (0.00042s latency).
ot shown: 977 closed tcp ports (reset)
ORT     STATE SERVICE     VERSION
1/tcp   open  ftp         vsftpd 2.3.4
_ftp-anon: Anonymous FTP login allowed (FTP code 230)
 ftp-syst:
   STAT:
 FTP server status:
      Connected to 192.168.1.10
      Logged in as ftp
      TYPE: ASCII
      No session bandwidth limit
      Session timeout in seconds is 300
      Control connection is plain text
      Data connections will be plain text
      vsFTPd 2.3.4 - secure, fast, stable
_End of status
2/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
 ssh-hostkey:
   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
3/tcp   open  telnet      Linux telnetd
5/tcp   open  smtp        Postfix smtpd
 ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/
 Not valid before: 2010-03-17T14:07:45
_Not valid after:  2010-04-16T14:07:45
_ssl-date: 2026-03-14T13:36:09+00:00; +2s from scanner time.
 sslv2:
   SSLv2 supported
```

5. Task 2 — Service Version Detection

Service version detection helps identify **software versions running on open ports**, which is important for vulnerability analysis.

Command used:

```
sudo nmap -sV -iL resolved.txt
```

Explanation

- **-sV** → Detects service versions

Result

The scan successfully identified service versions such as:

- Web servers
- SSH services
- Other running network services

```
┌──(kali㉿kali)-[~]
└─$ cat service_scan.txt
# Nmap 7.95 scan initiated Thu Mar 12 08:03:46 2026 as: /usr/lib/nmap/nmap -sV -i
Warning: 198.148.81.146 giving up on port because retransmission cap hit (10).
Nmap scan report for www.hackthissite.org (137.74.187.104)
Host is up (0.14s latency).
Other addresses for www.hackthissite.org (not scanned): 137.74.187.103 137.74.187
rDNS record for 137.74.187.104: hackthissite.org
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE  SERVICE    VERSION
22/tcp    closed ssh
80/tcp    open   tcpwrapped
443/tcp   open   tcpwrapped
9595/tcp  closed pds
16018/tcp closed unknown

Nmap scan report for 0-v3stage-cdn.hackthissite.org (198.148.81.146)
Host is up (0.0015s latency).
rDNS record for 198.148.81.146: customer.sharktech.net
Not shown: 705 filtered tcp ports (no-response), 293 closed tcp ports (reset)
PORT     STATE SERVICE    VERSION
80/tcp   open  tcpwrapped
8800/tcp open  tcpwrapped

Nmap scan report for qdb.hackthissite.org (137.74.187.153)
Host is up (0.017s latency).
rDNS record for 137.74.187.153: www.irc.hackthissite.org
Not shown: 976 filtered tcp ports (no-response)
PORT      STATE  SERVICE        VERSION
26/tcp    closed rsftp
80/tcp    open   tcpwrapped
443/tcp   open   tcpwrapped
808/tcp   closed ccproxy-http
```

6. Task 3 — Leveraging Nmap Scripting Engine (NSE)

Nmap includes a powerful feature called the **Nmap Scripting Engine (NSE)** that allows automated scripts to perform vulnerability detection and deeper enumeration.

6.1 Discovery Scripts

Discovery scripts collect additional information about the target system.

Command used:

```
nmap --script discovery 10.0.2.15 -oN NSE/discovery.txt
```

Result

The script gathered useful information about the target services and saved the results in:

```
discovery.txt
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV --script discovery,safe 192.168.1.9 -oN nse_discovery.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-14 09:52 EDT
No profinet devices in the subnet
Pre-scan script results:
| targets-ipv6-multicast-invalid-dst:
|   IP: fe80::a00:27ff:fe83:86e2  MAC: 08:00:27:83:86:e2  IFACE: eth0
|_  Use --script-args=newtargets to add the results as targets
| broadcast-dhcp-discover:
|   Response 1 of 1:
|     Interface: eth0
|     IP Offered: 192.168.1.11
|     Server Identifier: 192.168.1.1
|     Subnet Mask: 255.255.255.0
|     Router: 192.168.1.1
|     Domain Name Server: 192.168.1.1
|_    NetBIOS Name Server: 192.168.1.1, 192.168.1.1
| ipv6-multicast-mld-list:
|   fe80::a00:27ff:fe83:86e2:
|     device: eth0
|     mac: 08:00:27:83:86:e2
|     multicast_ips:
|       ff02::1:ff83:86e2         (NDP Solicited-node)
|   fe80::a266:acbe:9015:1cab:
|     device: eth0
|     mac: a4:fc:77:75:a7:d1
|     multicast_ips:
|       ff02::1:ff15:1cab         (NDP Solicited-node)
|       ff02::1:ffea:e539         (Solicited-Node Address)
|       ff02::1:ff1e:94fa         (Solicited-Node Address)
|       ff02::1:3                 (Link-local Multicast Name Resolution)
```

6.2 Vulnerability Scripts

Vulnerability scripts help identify potential security weaknesses in services.

Command used:

```
nmap --script vuln 10.0.2.15 -oN vuln_scanning.txt
```

Result

The scan highlighted possible vulnerabilities on the Metasploitable2 machine.

```
| ssl-dh-params:
|   VULNERABLE:
|   Diffie-Hellman Key Exchange Insufficient Group Strength
|     State: VULNERABLE
|       Transport Layer Security (TLS) services that use Diffie-Hellman groups
|       of insufficient strength, especially those using one of a few commonly
|       shared groups, may be susceptible to passive eavesdropping attacks.
|     Check results:
|       WEAK DH GROUP 1
|             Cipher Suite: TLS_DHE_RSA_WITH_AES_128_CBC_SHA
|             Modulus Type: Safe prime
|             Modulus Source: Unknown/Custom-generated
|             Modulus Length: 1024
|             Generator Length: 8
|             Public Key Length: 1024
|     References:
|_      https://weakdh.org
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
|_irc-unrealircd-backdoor: Looks like trojaned version of unrealircd. See http://seclists.org/fulldisclosure/2010/Jun/277
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:83:86:E2 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
|_smb-vuln-ms10-054: false
|_smb-vuln-ms10-061: false
```

7. Task 4 — Stealth & Firewall Evasion Techniques

Stealth scanning techniques are used to **avoid detection from firewalls, IDS, or IPS systems**.

These techniques help penetration testers perform scans more quietly.

7.1 Packet Fragmentation

Packet fragmentation breaks scan packets into smaller pieces to bypass security detection systems.

Command used:

```
sudo nmap -f 10.0.2.15
```

Result

The fragmented packets allowed the scan to proceed while reducing the chance of detection.

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -f 192.168.1.6
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-14 14:06 EDT
Nmap scan report for 192.168.1.6
Host is up (0.00035s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
```

7.2 Decoy Scanning

Decoy scanning hides the attacker's real IP address by sending scan packets from multiple spoofed IP addresses.

Command used:

```
sudo nmap -D RND:10 10.0.2.15
```

Explanation

- **-D** → Decoy scanning
- **RND:10** → Generates 10 random decoy IP addresses

Result

This technique made it difficult for the target system to determine the actual source of the scan.

```
┌──(kali㉿kali)-[~]
└─$ sudo nmap -D RND:5 192.168.1.6
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-14 14:09 EDT
Nmap scan report for 192.168.1.6
Host is up (0.00063s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:83:86:E2 (PCS Systemtechnik/Oracle VirtualBo
```

8. Key Learning Outcomes

During this week, I gained practical experience in:

- Network scanning using Nmap
- TCP and UDP port scanning
- Service and version detection
- Using the Nmap Scripting Engine (NSE)
- Implementing stealth scanning techniques
- Understanding how attackers perform reconnaissance

These skills are fundamental for **ethical hacking, penetration testing, and red teaming operations**.

9. Conclusion

This week's internship activities provided hands-on experience in **network reconnaissance and scanning methodologies**. Using Nmap, I was able to identify active hosts, open ports, running services, and potential vulnerabilities.

The exercises also demonstrated the importance of stealth techniques and automation using NSE scripts in modern penetration testing.

This knowledge forms a critical foundation for performing advanced security assessments in real-world environments.